

Evolve — Full Project Audit Report

React Native 0.80.2 CLI · Firebase Auth / Firestore / Storage / FCM / Functions
Read-only review · July 14, 2026 · No code changes made during this audit

Not store-ready

4 critical blockers

~293 TS files · ~39k LOC

6.0 / 10

Project health

3.5 / 10

Production readiness

No

Android deploy ready

No

iOS deploy ready

Deployment verdict. Neither Play Store nor App Store submission is viable yet. Blockers: Firestore self-promotion to admin on profile create, committed YouTube API key, Android release signed with the debug keystore, client-trusted exam/quiz scoring with answer keys readable by learners, iOS sample bundle ID, and empty iOS App Icon assets.

1. Architecture snapshot

Layered **presentation → services → domain** with React Context auth (no Redux/Zustand). Custom home tab bar; Admin nested stack gated client-side. Content lives in Firestore subscriptions via HomeFeedContext (~8 live listeners when Main is active).

Strengths: typed domain models, signup Auth rollback, account deletion, allowBackup=false, admin content writes mostly rule-gated, structured appAlert UX.

Gaps: security rules holes, almost no tests, no Error Boundaries / Crashlytics, docs still RN template.

2. Findings by severity

Critical: 4 · High: 10 · Medium: 12 · Low: 8 · Suggestions: 9

3. Critical

Issue	Location	Recommended fix
Any signed-in user can create users/{uid} with role: admin and unlock CMS + admin callables	firestore.rules:8–13, 717–718	Force create role == 'user'; promote admins only via Admin SDK / Cloud Function; audit existing admins
YouTube API key committed; .env not in .gitignore	.env:4 · .gitignore · config/youtube.ts:9	Rotate/restrict key; gitignore .env; consider proxy via Cloud Functions
Release APK/AAB signed with debug keystore	android/app/build.gradle:102–105	Create release keystore; gitignore secrets; wire signingConfigs.release
Exam/quiz answer keys readable; scores written by client without server verification	firestore.rules:107–118, 584–625 · ExamAttemptScreen ~180 · QuizAttemptScreen ~211	Hide correctChoiceIndex from learners; score via callable Function

4. High

Issue	Location	Recommended fix
Storage CMS paths writable by any auth user under their uid (abuse / cost)	<code>storage.rules:22–81</code>	Require admin for thumbnails/PDFs; keep only <code>profileImages</code> user-writable
iOS still uses sample Bundle ID; mismatch with Android <code>com.evolve</code>	<code>project.pbxproj:294,321</code> · <code>GoogleService-Info.plist:15–16</code>	Set production Bundle ID; regenerate Firebase/Google Sign-In configs
iOS App Icon set has no image files — App Store reject	<code>Images.xcassets/AppIcon.appiconset/</code>	Add full icon set including 1024×1024
iOS Push incomplete: background mode only, no entitlements / APNs	<code>Info.plist:49–52</code> · <code>AppDelegate</code> · <code>project.pbxproj</code>	Enable Push capability, <code>aps-environment</code> , APNs key in Firebase
Empty <code>CLLocationWhenInUseUsageDescription</code> (unused; review risk)	<code>Info.plist:45–46</code>	Remove key if unused
Privacy Policy / Terms rows non-functional (no <code>onPress</code>)	<code>SettingsScreen.tsx:154–168</code>	Link to hosted legal URLs via <code>Linking.openURL</code>
Unpublished content + answer keys only filtered client-side	<code>firestore.rules</code> reads · <code>examsService</code> filter	Rules: <code>allow read if isAdmin() isPublished == true</code> ; strip answers
Stale admin role possible via <code>isRicherUserProfile</code> (ignores role)	<code>mapUserProfile.ts:81–100</code> · <code>AuthContext</code> hydration	Always apply remote role; include role in richness compare
Google Sign-In does not provision Firestore profile	<code>googleSignInService.ts:79–118</code>	Mirror email signup: <code>createUserProfileIfNotExists</code> after Google auth
No React Error Boundary; no Crashlytics/Sentry	<code>App.tsx</code> · <code>package.json</code>	Add root <code>ErrorBoundary</code> + crash reporting before release

5. Medium

Issue	Location	Recommended fix
ProGuard/R8 disabled for release	<code>android/app/build.gradle:62,106–107</code>	Enable minify; add RN/Firebase keep rules
Version mismatch: native 1.0 vs Settings UI 2.1.1 vs package 0.0.1	<code>build.gradle</code> · <code>SettingsScreen</code> · <code>package.json</code>	Single source of truth for store versioning
<code>PrivacyInfo.xcprivacy</code> declares no collected data types	<code>ios/Evolve/PrivacyInfo.xcprivacy</code>	Declare email, name, phone, photos, user ID, etc.
Kids track collects PII without age gate / COPPA flow	Track kids professionals across app + rules	Age gate, parental consent, store age rating alignment
HomeFeed mounts many concurrent <code>onSnapshot</code> listeners	<code>HomeFeedContext.tsx</code>	Pause off-screen; coalesce; paginate notifications

Issue	Location	Recommended fix
Password policy mismatch: UI min 8 vs Firebase msg mentions 6	SignUpScreen · signUpService.ts:29–30	Align messages and Auth password policy
Admin stack omitted while roleLoading (nav race)	MainStack.tsx:43,106–108	Defer admin entry until role resolved; keep Unauthorized fallback
Placeholder Android icons; no adaptive icon; default splash	android res/mipmap · styles.xml · LaunchScreen	Brand adaptive icons + branded splash
Hardcoded Google Web Client ID + support contacts in source	config/googleSignIn.ts · config/support.ts	Env/flavor-driven configs for prod vs staging
README is mostly RN template; no architecture / deploy / secrets docs	README.md	Document env setup, Firebase deploy, signing, roles
Test coverage util-only (4 test files)	__tests__/ · utils tests	Auth gating, rules, scoring, critical services tests
App Navigation root stack untyped	AppNavigation.tsx:27	Type createNativeStackNavigator with RootStackParamList

6. Low

Issue	Location	Recommended fix
Deprecated ProfileDisplayProvider / useUserProfile still present	ProfileDisplayContext · hooks/useUserProfile	Remove dead aliases
store/ is types-only — confusing name	src/store/	Rename to types/ or models/
Custom tabs outside React Navigation (no deep-linkable tab state)	HomeBottomTabBar · HomeScreen	Adopt @react-navigation/bottom-tabs when deep links matter
Debug cleartext traffic enabled (debug flavor only)	debug AndroidManifest	Keep out of release (already OK)
ATS allows local networking	Info.plist:38–44	Acceptable for Metro; ensure release is HTTPS-only
Unguarded console.warn in some admin save paths	ManageContinueLearningPlaylists ~323	Gate with __DEV__
react-native-webview PDF via Google Docs viewer	pdfViewerUrl util	Only load trusted admin URLs; consider native PDF
No custom App Links / universal links for content	AndroidManifest · Info.plist	Add if marketing / share flows need them

7. Suggestions

Idea	Why	Priority
Add Analytics + Crashlytics (or Sentry)	No production crash/telemetry today	P0 before soft launch
Offline / cache strategy for courses & home feed	Many live listeners; weak offline UX	P1

Idea	Why	Priority
Server-driven remote config for feature flags	Admin CMS + quiz competitions benefit	P2
E2E (Detox/Maestro) for auth → home → exam flow	Only unit smoke tests today	P1
Move YouTube playlist fetch to Cloud Functions	Keep API key off device	P0 with secrets hygiene
Image CDN / resizing for profile & CMS uploads	5MB uploads; list performance	P2
Accessibility audit (labels, contrast, Dynamic Type)	Educational / kids audience	P1
CI: lint, typecheck, unit tests, rules tests on PR	No automated quality gate observed	P1
Play Integrity / App Check	Reduce spoofed clients abusing rules holes	P1 after rules fix

8. Platform & store readiness

Android (Play) — Blocked

Package `com.evolve` · minSdk 24 · targetSdk 35 · versionCode 1 — SDK targets are fine. Blockers: debug signing on release, committed secrets, admin privilege escalation, quiz integrity, Privacy/Terms links for Data Safety / policy, brand assets. Permissions (INTERNET, POST_NOTIFICATIONS, READ_MEDIA_IMAGES) align with features; CAMERA not declared (gallery-only) — good. ProGuard/R8 still disabled.

iOS (App Store) — Blocked

Bundle ID still `org.reactjs.native.example.Evolve`. Empty App Icons. Push incomplete. Empty location string. Incomplete Privacy Manifest data types. Signing team / entitlements not production-ready. Plus shared blockers: Firestore role create, secrets, legal links, kids/age compliance if under-13 content applies.

Policy compliance notes

- Account deletion exists (PrivacySettingsScreen + deleteAccountService) — required for both stores.
- Privacy Policy / Terms UI present but not wired — likely rejection / Data Safety failure.
- Kids track + PII without age gate risks COPPA / App Store kids category scrutiny.
- Declare collecting data accurately in Play Data Safety and PrivacyInfo.xcprivacy before submit.

9. Prioritized production roadmap

- **P0 — Security rules:** Lock Firestore create role to user; audit existing admins; tighten isPublished + answer-key reads; admin-only Storage CMS writes.
- **P0 — Secrets:** Rotate YouTube key; gitignore .env; move key server-side; scrub history if repo shared.
- **P0 — Quiz integrity:** Server-side exam/quiz scoring (callable); stop trusting client correctCount/xpEarned.
- **P0 — Packaging:** Android release signing; iOS production Bundle ID + Firebase/Google Sign-In rebuild; App Icons; remove empty location key.
- **P0 — Legal / compliance:** Wire Privacy Policy & Terms URLs; complete PrivacyInfo + Play Data Safety; age gate if kids track includes under-13.

- **P1 — Resilience:** Fix Google profile provisioning + stale role hydration; Error Boundary + Crashlytics; iOS Push entitlements/APNs.
- **P1 — Release polish:** Enable R8; brand Android adaptive icons/splash; align version numbers; CI lint/test/rules.
- **P2 — Quality:** Expand tests (auth, guards, scoring); HomeFeed listener optimization; accessibility; App Check; product docs.

Scores rationale. Health 6.0 reflects architecture maturity and feature depth. Production readiness 3.5 reflects ship-blocker density. Fixing the P0 security and packaging items would likely raise production readiness into the 7–8 range before soft launch.

Stack reviewed: RN 0.80.2 / React 19.1 / RNFB 24 / Reanimated 3 / Navigation 7 · Hermes + New Architecture enabled · Functions Node 20

Generated from Evolve audit · July 14, 2026 · Confidential engineering review